DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
COMPUTER SCIENCE DEPARTMENT

COURSE SYLLABUS

Second Semester, SY 2024-2025

COURSE CODE	**:**	S-CSPC224
COURSE TITLE	**:**	DISCRETE STRUCTURES 2
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	Three Units
LEARNING MODALITY	**:**	Traditional blended learning model
PRE-REQUISITES	**:**	S-CSPC212
CO-REQUISITES	**:**	None
PROFESSOR	**:**	Ms. Elsie G. Cabalsa
CONSULTATION HOURS	**:**	Monday 8:00 am to 10:00 am COS100-C
		Thursday 1:00 pm to 2:00 pm COS100-C
CONTACT INFORMATION	**:**	egcabalsa@dlsud.edu.ph (046) 4811900 local 3134

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course provides fundamental mathematical structure and logical principles that are important to computer science and develops a deeper understanding of the computing problems that require mapping to permutations, combinations of a set, modular arithmetic reasoning, and counting techniques. The course explains a set theory and advanced counting techniques, graphs, and complex applications to solve challenging problems related to the subject with real-world applications.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. demonstrate the application of discrete structures in different fields of computer science;
CLO2. evaluate Boolean functions and simplify expressions using properties of Boolean algebra;
CLO3. explain some of the concepts of number theory;
CLO4. analyze the fundamentals theorem of arithmetic;

CLO5. apply counting techniques in solving arithmetic and counting problems;
CLO6. understand graphs, trees, and their applications;
CLO7. apply mathematical arguments and reasoning in proving mathematical statements; and
CLO8. demonstrate 21st-century skills in all learning activities.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	**Topic Learning Outcomes**
CLO1. Demonstrate the application of discrete structures in different fields of computer science. CLO2. Evaluate Boolean functions and simplify expressions using properties of Boolean algebra.	TLO 1. Simplify and prove Boolean expressions. TLO 2. Convert Boolean expressions to logic gates and vice versa. TLO 3. Construct a logic circuit.
CLO3. Explain some of the concepts of number theory. CLO4. Analyze the fundamentals theorem of arithmetic.	TLO 4. Understand number theory. TLO 5. Know the number systems/integer representations. TLO 6. Understand modular arithmetic. TLO 7. Solve problems in primes and divisibility. TLO 8. Find the LCM and GCD.
CLO5. Apply counting techniques in solving arithmetic and counting problems.	TLO 9. Discuss the basics of counting. TLO 10. Understand permutation and Combinations as well as the Pigeon Principle.

Module No. and Learning Outcomes	**Week**	**Teaching-Learning Activities/Assessment Strategies**	**Technology Enabler**	**Onsite / F2F**	**Asynchronous**	**Allotted Hours**
Module 0 Module 1	Week 1 Jan. 20-Jan 25	Gospel reading, reflection, and prayer • Presentation of Module 0 - Course Overview • Getting-to-know: Teacher and Students • Presentation of Module 1 – Boolean Algebra: Boolean Functions • Class participation: Seatwork No. 1 10-item questions on the Truth table	MS PowerPoint, Multimedia Devices (Laptop, Projector)	✓		3.0
Module 1 CLO1 CLO2 TLO1 TLO2 TLO3	Week 2 Jan 27-Feb 1	• Gospel reading, reflection, and prayer • Presentation of Module 1 – Boolean Algebra o Boolean Expressions and Circuits • Enabling Assessment: Short Quiz No. 1 25-item quiz on Simplification and Logic Circuit	MS PowerPoint, Multimedia Devices (Laptop, Projector)	✓		3.0
Module 2 CLO3, TLO4 TLO5 TLO6 TLO7 TLO8	Week 3 Feb 3-Feb 8	• Gospel reading, reflection, and prayer • Module 2 – Number Theory (Self-paced Reading) o Logic Gates o Number Theory - Euclidean Algorithm • Enabling Assessment 2: Logic Gate in the World and Concept of Universal Gates	School Book		✓	3.0

Module 2 CLO3 TLO4 TLO5 TLO6 TLO7 TLO8	Week 4 Feb 10-Feb 15	• Gospel reading, reflection, and prayer • Continuation Presentation of Module 2 – Number Theory • Integer Representation • Constructing base expansion • Conversion Binary, Octal and Hexadecimal • Enabling Assessment: Short Quiz No. 2 Conversion Binary, Octal and Hexadecimal	MS PowerPoint, Multimedia Devices (Laptop, Projector)	✓		3.0
Module 3 CLO5 TLO9 TLO10	Week 5 Feb 17-Feb 22	• Gospel reading, reflection, and prayer • Enabling Assessment: Long Quiz No. 1 • Coverage: Modules 1 and 2 o 50-item long quiz on Boolean Algebra and Number Theory • Presentation of Module 3 – Counting o The Basics of Counting • Class Participation: 10 Items Questions, Basic Counting Principles	. MS PowerPoint, Multimedia Devices (Laptop, Projector)	✓		3.0
	Week 6 Feb 24-Feb Mar 1	• Gospel reading, reflection, and prayer • Module 3 – cont. Counting (Self-paced Reading) - Permutation and Combination - The Pigeonholes • Enabling Assessment: Seatwork No. 2 o 15-item activity on Permutation and Combination	School Book		✓	3.0
	Week 7 Mar 3- Mar 8	• Gospel reading, reflection, and prayer • Module 3 – Counting - The Pigeonholes • Self-paced Reading	School Book		✓	3.0
	Week 8 Mar 10- Mar 15	• Gospel reading, reflection, and prayer • Review Week in preparation for the Midterm Examination • Self-Paced Reading	MS Teams Schoolbook		✓	3.0
Week 9 Mar 17 – Mar 22		**MIDTERM EXAMINATION** Covering Modules 1 to 3 Pen and Paper				3.0
TOTAL				12	12	27.00

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Demonstrate the application of discrete structures in different fields of computer science. CLO6. Understand graphs, trees, and their applications.	TLO 11. Understand basic terminologies in graphs and trees. TLO 12. Know directed and undirected graphs. TLO 13. Understand the applications of trees.
CLO7. Apply mathematical arguments and reasoning in proving mathematical statements.	TLO 14. Define the two-step process for mathematical induction. TLO 15. Used mathematical induction to prove statements.

CLO8. Demonstrate 21st-century skills in all learning activities.	*Collaboration, Skilled Communication, Knowledge Construction, Self-regulation, Real-World Problem-Solving, Use of ICT (not part of the topic, but integrated into assessments)*

Module No and Learning Outcomes	**Week**	**Teaching-Learning Activities / Assessment Strategies**	**Technology Enabler**	**Onsite / F2F**	**Asynchronous**	**Allotted Hours**
Module 4 CLO1 CLO6 TLO11 TLO12 TLO13	Week 10 Mar 24 – Mar 28	• Gospel reading, reflection, and prayer • Module 4: Graphs o Basic Terminologies o Directed graphs o Undirected graphs • Self-paced Reading *Sustainability Week (Online Class)*	School Book		✓	
	Week 11 Mar 31- Apr 5	• Gospel reading, reflection, and prayer • Presentation of Module 4: Graphs o Basic Terminologies o Directed graphs o Undirected graphs • Enabling Assessment: 20-item quiz on different types and classes of graphs	MS PowerPoint, Multimedia Devices (Laptop, Projector)	✓		3.0
	Week 12 Apr 7-Apr 12	• Gospel reading, reflection, and prayer • Presentation of Module 4: Continuation o Basic Terminologies o Directed graphs o Undirected graphs • Enabling Assessment: Group Activity Collaborative Graph Construction	MS PowerPoint, Multimedia Devices (Laptop, Projector)	✓		3.0
Module 5 CLO7 TLO14 TLO15	Week 13 Apr 12-Apr 19	• Gospel reading, reflection, and prayer • Presentation of Module 5: Mathematical Induction and Reasoning o Introduction to Mathematical Induction o Strong Induction and Well-ordering o Recursive Algorithms INDEPENDENT LEARNING WEEK HOLY WEEK	MS PowerPoint, Multimedia Devices (Laptop, Projector)		✓	3.0
Module 4 Module 5 CLO5 CLO6 TLO11 TLO12 TLO13	Week 14 Apr 21-Apr 26	• Gospel reading, reflection, and prayer • Module 6 – Trees (Self-paced reading) o Introduction to Trees • Enabling Assessment: Seatwork No. 4 • 10-item questions on identifying trees RESEARCH WEEK	School Book		✓	3.0

Module 6 CLO1 CLO6 TLO11 TLO12 TLO13	Week 15 Apr 28-May 3	• Gospel reading, reflection, and prayer • Enabling Assessment: Long Quiz No. 2 • Coverage: Modules 4 and 5 • 50-item questions on Graphs and Mathematical Induction	MS PowerPoint, Multimedia Devices (Laptop, Projector)	✓		3.0
	Week 16 May 5-May 10	• Gospel reading, reflection, and prayer • Presentation of Module 6: Trees ○ Applications of Trees ○ Tree Traversal • Enabling Assessment: Seatwork No. 5 • (15-item questions on Application and Tree Traversal and their subtree)	MS PowerPoint, Multimedia Devices (Laptop, Projector)	✓		3.0
	Week 17 May 12-May 17	• Gospel reading, reflection, and prayer • Review Week in preparation for the Final Examination • Self-Paced Reading			✓	
Week 18 May 19-May 24	**FINAL EXAMINATION** Covering Modules 4 to 6 Pen and Paper					3.0
			TOTAL	12.0	12.0	27.0

GRADING SYSTEM:

Midterm	%	Final	%
Enabling Assessments	50%	Enabling Assessments	50%
Summative Assessment	30%	Summative Assessment	30%
Class Participation	20%	Class Participation	20%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
   a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
   b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
      i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.

ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through the Schoolbook messaging feature.
iii. Email – you may email your professor. (Professor's email will be given during the course orientation)

c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.
   c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
   a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
   b. Using any illegally obtained software is strictly prohibited.
   c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
   d. Giving of passwords to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 accounts other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
   a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
   b. Students must always be honest; cheating and plagiarism in any form will merit a grade of 0.00.
   c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones cannot also be used as calculators during examinations.
   d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

On-Site References

Provider	Reference Material
AEA Lib QA 9. K435 2024	Khoury J. 2024. **Tale of Discrete Mathematics, A: A Journey Through Logic, Reasoning, Structure and Graph Theory.** Singapore: World Scientific Publishing Co. Pte Ltd. LLC.
AEA Lib QA 9. D429 2020	Rosenberg AL, Trystran D. 2020. **Understanding Mathematics, Understanding Computing: Discrete Mathematics That All Computing Students Should Know.** Massachusetts (USA): Springer LLC.
AEA Lib QA 39.3 .D631m 2017	Borres M. 2017. **Discrete Mathematics: An Introduction.** New York (NY): Arcler Press LLC.
AEA Lib QA 39.3 .Ad95 2016	Tedford S, et al. 2016. **Advanced Discrete Mathematics.** New York (NY): Magnum Publishing LLC.

Online References:

Call Number/ E-provider	Reference Material

e-book dlsu.355658	Rosen KH. *Discrete Mathematics and Its Application. 8th ed*. McGraw Hill Higher Education; accessed 2019 May 5. Available from: *https://aea.dlsud.edu.ph/digitalresource/1648701456743_Discrete_Mathematics_and_Its_Applications_(TOC)_-_Rosen.pdf*
Journal	Balla R, Padley R. 2019. **Advances in Discrete Mathematics: From Combinatorics to Cryptography.** accessed 2019 Oct 22. Available from: https://turcomat.org/index.php/turkbilmat/article/view/14624

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2nd Semester 2024-2025



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